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High-order optical nonlinearity at low light levels

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received 16 January 2012; accepted in final form 20 March 2012

published online 25 April 2012

PACS 42.65.-k – Nonlinear optics

PACS 37.10.Jk – Atoms in optical lattices

PACS 37.10.Vz – Mechanical effects of light on atoms, molecules, and ions

Abstract – We observe a nonlinear optical process in a gas of cold atoms that simultaneously displays the largest reported fifth-order nonlinear susceptibility $\chi^{(5)} = 1.9 \times 10^{-12} \text{ (m/V)}^4$ and high transparency. The nonlinearity results from the simultaneous cooling and crystallization of the gas, and gives rise to efficient Bragg scattering in the form of six-wave mixing at low light levels. For large atom-photon coupling strengths, the back-action of the scattered fields influences the light-matter dynamics. We confirm this interpretation by investigating the nonlinearity for different polarization configurations. In addition, we demonstrate excellent agreement between our experimental measurements and a theoretical model with no free parameters, and compare our results to those obtained using alternative approaches. This system may have important applications in many-body physics, quantum information processing, and multidimensional soliton formation.

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Since the first observation of second-harmonic generation of a ruby laser beam over 50 years ago, there has been sustained effort to improve the nonlinear optical (NLO) interaction strength of materials. One ultimate goal is to realize nonlinear interactions at the single-photon level, which will lower the operating power of devices. There is also a growing need for single-photon nonlinearities for quantum information applications [1].

The strength of a material's nonlinear optical response is characterized by the n -th order susceptibility tensor $\chi^{(n)}$, which relates the nonlinear polarization of the material $\vec{P}^{NL}$ to the electric-field strength $\vec{E}$ via $\vec{P}^{NL} = \epsilon_0 [\chi^{(3)} : \vec{E}\vec{E}\vec{E} + \chi^{(5)} : \vec{E}\vec{E}\vec{E}\vec{E}\vec{E} + \dots]$ for an isotropic material [2]. Because the magnitude of the susceptibility typically decreases with increasing order, most low-light-level studies to date have focused on lower-order processes. For example, there have been numerous observations of low-light-level NLO interactions based on third-order ($\chi^{(3)}$) processes created via electromagnetically induced transparency (EIT) [3–6], where a strong coupling beam creates a quantum interference effect that simultaneously renders the medium nearly transparent while enhancing the nonlinearity [7]. Other recent observations of strong third-order NLO effects include a two-photon absorptive switch actuated using <20 photons interacting with atoms

in a hollow-core photonic bandgap fiber [8] and an optical pattern-based switch using ~ 600 photons interacting with a warm atomic vapor [9].

Despite the success of these approaches, some applications require or can benefit from higher-order nonlinearities. Materials with a large, fifth-order ($\chi^{(5)}$) response, for example, can lead to new phenomena, such as liquid light condensates [10] and transverse pattern and soliton formation [11], and play an important role in quantum information networks through enabling 3-qubit quantum processing [12,13], providing new sources of correlated pulse pairs [14], and acting as quantum memories [14,15]. Quintic media can also improve high-precision measurements [16] and reduce phase noise for enhanced interferometry performance [17]. Thus, the realization of efficient $\chi^{(5)}$ materials is important for fundamental studies in quantum nonlinear optics as well as improving the performance of NLO devices.

In this letter, we report the discovery of a *dissipation-enhanced* NLO process that gives rise to the largest fifth-order ($\chi^{(5)}$) NLO susceptibility ever reported while simultaneously having high transparency. Our NLO material consists of a gas of cold atoms initially in thermal equilibrium, which is illuminated by weak, frequency-degenerate laser beams (frequency ω). For certain polarization configurations, the light fields (both applied and self-generated via wave mixing) act on the atomic center-of-mass motion

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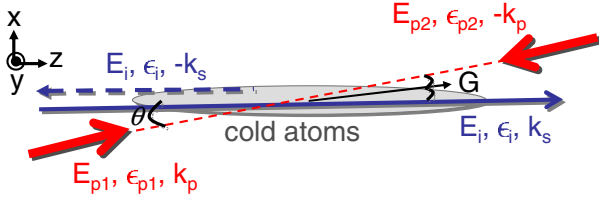


Fig. 1: (Color online) Schematic of the experimental setup. We take $z = 0$ ($z = L$) to be the left (right) side of the cloud.

Table 1: Polarization configurations for the pump and signal beams.

	$\aleph_{\parallel}$	$\aleph_{\perp}$	$\Sigma_{\parallel}$	$\Sigma_{\perp}$	$\Xi_{\parallel}$	$\Xi_{\perp}$	$\Theta_{\parallel}$	$\Theta_{\perp}$
$\hat{\epsilon}_{p1}$	$\uparrow\downarrow$	$\uparrow\downarrow$	$\uparrow\downarrow$	$\uparrow\downarrow$	$\odot$	$\odot$	$\odot$	$\odot$
$\hat{\epsilon}_{p2}$	$\leftrightarrow$	$\leftrightarrow$	$\uparrow\downarrow$	$\uparrow\downarrow$	$\odot$	$\odot$	$\odot$	$\odot$
$\hat{\epsilon}_s$	$\uparrow\downarrow$	$\leftrightarrow$	$\uparrow\downarrow$	$\leftrightarrow$	$\odot$	$\odot$	$\odot$	$\odot$

to cool and crystallize the gas and lead to Bragg scattering via the generated atomic density gratings. This cooling arises via the dissipative Sisyphus force [18] and occurs efficiently (*i.e.*, requires the scattering of only tens of photons per atom) so that absorption can be made small. We present additional support for this interpretation by demonstrating qualitatively similar results for a variety of polarization configurations that allow for atomic cooling and localization. In addition, we present a model for one polarization configuration that describes well the experimental results with no free parameters.

In contrast to previous studies of wave mixing via atomic bunching that produce a third-order NLO response [19–21], the inclusion of dissipative effects in our system cause the lowest-order nonlinearity to be fifth order in the applied fields. Surprisingly, the achievable nonlinear interaction strength observed in our experiments from the $\chi^{(5)}$ response is just as large as that obtained in previous experiments dominated by a $\chi^{(3)}$ response. This strong light-matter coupling enables the scattered fields to act back on the vapor, which suggests that our system may be particularly interesting for studies of many-body physics with long-range interactions [22].

To make our discussion concrete, we consider the situation shown in fig. 1, where optical fields interact with a cloud of cold atoms in a pencil-shaped geometry. A pair of intensity-balanced, counterpropagating pump fields (intensity I_p , wave vectors $\pm\vec{k}_p$, polarizations $\epsilon_{p1,p2}$) are inclined by an angle $\theta = 10^\circ$ relative to the cloud’s long axis. A weak signal field (I_s , $\vec{k}_s$, ϵ_s) is injected along the z -axis and couples with the pump fields to generate a counterpropagating idler field (I_i , $-\vec{k}_s$, ϵ_i) via NLO wave mixing. Table 1 defines our notation for the pump and signal beam polarizations in the linearly and circularly polarized bases.

In our experiment, we use an anisotropic magneto-optical trap (MOT) to confine ^{87}Rb atoms in the

$5^2S_{1/2}(F=2)$ state within a cylindrical region of length $L = 3\text{ cm}$ (along $\hat{z}$) and diameter $W = 300\text{ }\mu\text{m}$ (along $\hat{x}$ and $\hat{y}$) [23]. This configuration enables us to achieve typical atomic densities of $\sim 5 \times 10^{10}\text{ cm}^{-3}$ for atoms isotropically cooled to $T_{eq} = 20\text{--}30\text{ }\mu\text{K}$. The pump and probe beams are derived from the same laser, are detuned from the $5^2S_{1/2}(F=2) \rightarrow 5^2P_{3/2}(F'=3)$ transition (transition frequency ω_{23}) by $|\Delta|/\Gamma = 3\text{--}20$ ($\Delta = \omega - \omega_{23}$) and have diameters of 3 mm and $200\text{ }\mu\text{m}$, respectively. We use pump beam intensities of up to a few mW/cm^2 and a signal beam intensity of $3\text{ }\mu\text{W}/\text{cm}^2$, which we control via acousto-optic modulators with response times of $< 100\text{ ns}$. Thus, all beams are well below the off-resonance electronic saturation intensity $I_{sat}^\Delta = I_{sat}[1 + (2\Delta/\Gamma)^2]$, where $I_{sat} = 4\epsilon_0 c \hbar^2 / (\mu^2 \Gamma^2) = 1.6\text{ mW}/\text{cm}^2$ is the resonant saturation intensity, $\Gamma/2\pi = 6\text{ MHz}$ is the natural transition linewidth, $\mu = 2.53 \times 10^{-29}\text{ C}\cdot\text{m}$ is the reduced transition dipole moment, ϵ_0 is the permittivity of free space, and c is the speed of light in vacuum.

To measure the system’s NLO response, we cycle between a wave-mixing and a cooling and trapping phase. During the wave-mixing phase, we turn on only the pump and signal beams for $\sim 1\text{ ms}$ and record the time-dependent intensities of the signal ($I_s(z=L)$) and idler ($I_i(z=0)$) beams as they exit the cloud. After $\sim 200\text{ }\mu\text{s}$, the system reaches a steady state that persists for $\sim 1\text{ ms}$ (at which time expansion of the cloud in the y -direction reduces the density and, consequently, the nonlinearity). We then cool and trap the atoms with only the MOT beams for 99 ms and repeat the cycle. We have verified that the MOT magnetic fields, which remain on during the experiment, do not affect the NLO response [24].

We look first at the $\aleph_{\perp}$ configuration because it has been well studied previously and, as we show, leads to the largest NLO response. Figure 2(a) shows the steady-state reflectivity $R = I_i(0)/I_s(0)$ as a function of I_p for different Δ , where the points (solid curves) correspond to experimental measurements (theoretical predictions, which we discuss later). The reflectivity scales superlinearly with I_p and can approach 1 for weak pump intensities (*i.e.*, $I_p \ll I_{sat}$). Beyond $R \sim 1$, an optical instability occurs that gives rise to self-generated signal and idler fields in the absence of an injected signal beam [24]. This instability, coupled with the rapid variation of R on I_p , ultimately limits the largest values of R that we can measure.

For a fixed I_p , R decreases with increasing $|\Delta|$. Nevertheless, this decrease occurs sufficiently slowly that we can still obtain large R while operating far from resonance (where absorption is minimal). Figure 2(b) shows the signal-beam transmission $T = I_s(L)/I_s(0)$ in the absence of the pump beams, given by $T_0 = \exp(-\alpha_\Delta L)$, where $\alpha_\Delta = \eta \hbar \omega \Gamma / 2 I_{sat}^\Delta$ is the off-resonance absorption coefficient and η is the average atomic density. For all of the detunings shown in fig. 2(a), $T_0 > 0.65$ and approaches 1 for the larger detunings. Thus, this light-matter interaction allows us to realize large nonlinearities with high

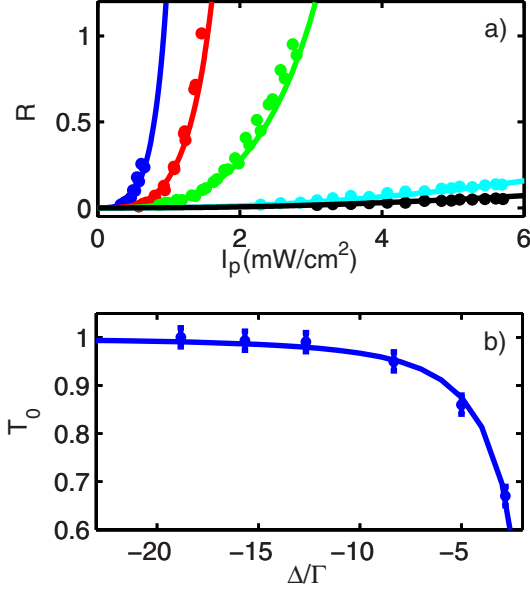


Fig. 2: (Color online) (a) Dependence of R on I_p for $\Delta/\Gamma = -3, -5, -7.3, -12.7$, and -18.8 (from left to right, respectively). (b) Signal-beam transmission for $I_p = 0$. For both figures, $\alpha_{\Delta=0}L = 13.4(\pm 0.5)$ (uncertainty corresponds to one standard deviation), we use the $\mathbb{N}\perp$ configuration, the points correspond to the experimental data, and the solid curves correspond to the theory via eqs. (5) and (4).

transparency for low input intensities (*e.g.*, we measure $R = 1$ for $I_p = 1.5 \text{ mW}/\text{cm}^2$ by working at $\Delta = -5\Gamma$ where $T_0 = 0.87$).

We interpret the mechanism underlying the probe beam amplification as Bragg scattering of pump photons into the probe beam direction via an atomic density grating. The forces resulting from the interference of a pump and nearly counterpropagating probe beam generate an atomic density modulation along $\vec{G} = \vec{k}_p + \vec{k}_s$ that is phase-matched for Bragg scattering. The resulting nonlinear atomic polarization is $\vec{P}_{s,i}^{NL} = \eta\mu^2 b \vec{E}_{p2,p1}/\hbar\Delta$, where b is the amplitude of the density modulation along $\hat{G}$, $\vec{E}_{p1,p2}$ are the pump electric fields, and $|\vec{E}_{p1}| = |\vec{E}_{p2}| = E_p$ [25]. Thermal motion limits the degree to which atoms can be localized along $\hat{G}$ so that lower temperatures result in larger values of b and, consequently, higher Bragg scattering efficiencies.

Because of this connection between atomic temperature and NLO scattering efficiency, we find that the dissipative optical lattice formed by the pump beams plays an important role in the nonlinearity (in contrast to other works in which the pump-beam lattice does not directly enhance the light-matter coupling [26,27]). The pump-beam lattice causes Sisyphus cooling, which transforms the atomic momentum distribution along $\hat{k}_p$ from a Maxwell-Boltzmann distribution to one that is well described by a double-Gaussian function [28,29]. One interprets the gas as a nonthermal system consisting

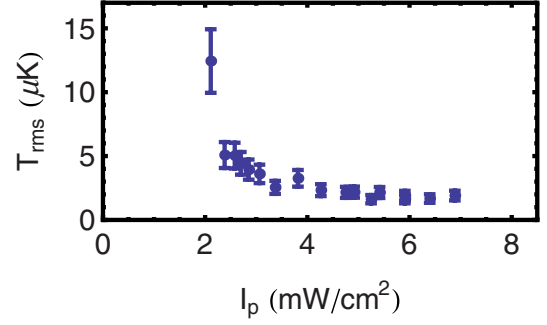


Fig. 3: (Color online) Atomic temperature as a function of I_p for $\Delta/\Gamma = -10$ in the $\mathbb{N}\perp$ configuration. The décrochage intensity corresponds to $I_d \sim 3 \text{ mW}/\text{cm}^2$.

of a cold component of atoms well localized in the pump-pump lattice at a temperature T'_c and an unbound, hot component at temperature T'_h undergoing anomalous diffusion. The cooling process transfers atoms from the hot to the cold component, where the fraction of atoms in each component is $f_{h,c}$, respectively.

We solve for $f_{c,h}$ by calculating numerically the steady-state momentum distribution for a $J_g = 1/2 \rightarrow J_e = 3/2$ transition using a Bloch state approach [30] and find that f_c initially increases linearly with I_p before asymptoting to 1 [24]. While no analytic solution for f_c exists, the simple functional form $f_c = 1 - f_h \cong \tanh(I_p/I_c)$ fits the numerical results well, where the characteristic intensity for the cooling process (and, therefore, the nonlinearity) is I_c . The characteristic intensity is directly proportional to and of the same order as the so-called décrochage intensity I_d , which is the intensity where the root-mean-squared (r.m.s.) width of the momentum distribution is a minimum (*i.e.*, where cooling optimally balances diffusive heating) [18]. For the chosen form of $f_{c,h}$, we predict that $I_c \cong I_d/2$. Perhaps surprisingly, both the cooling and heating rates vary with detuning in such a way that I_d and, therefore, I_c is independent of Δ [31]. As we discuss later, the independence of I_c on Δ plays an important role in achieving large NLO coupling strengths in our system.

We determine the value of I_d experimentally by measuring the r.m.s. temperature T_{rms} along $\hat{G}$ using recoil-induced resonance velocimetry [32]. Figure 3 shows that T_{rms} decreases rapidly for small I_p up to $I_p \sim 3 \text{ mW}/\text{cm}^2$ and then roughly levels off at $T_{rms} \cong 2 \mu\text{K}$. This trend is consistent with cooling in a $\text{lin}\perp\text{lin}$ lattice in the vicinity of I_d . From this measurement, we find that $I_d \sim 3 \text{ mW}/\text{cm}^2$, which agrees with the value of the décrochage intensity observed in previous studies using a 3D $\text{lin}\perp\text{lin}$ lattice to within a factor of 4 [31,33]. We attribute this difference to the fact that we work with a 1D lattice [28] and to the different scattering properties of our anisotropic MOT compared to spherical MOTs [34]. This value of I_d implies that the characteristic intensity $I_c \cong I_d/2 = 1.5 \text{ mW}/\text{cm}^2$, which agrees well with the value measured independently via monitoring the temporal decay of the

nonlinear response [24]. We can therefore access NLO interaction strengths comparable to those of resonantly-driven atoms while the atoms are nearly transparent, since $I_c \cong I_{sat}$ even for $|\Delta/\Gamma| \gg 1$.

The dissipative pump-beam lattice therefore assists in cooling and loading atoms into the lattice along $\hat{G}$, which is phase-matched for scattering pump light into the probe beam directions. For weak bunching and $(2\Delta/\Gamma)^2 \gg 1$, we find that the amplitude of the phase-matched density grating is given by [35]

$$b = \frac{\hbar\Gamma}{4(\Delta/\Gamma)} \frac{\epsilon_0 c(E_s E_{p2}^* + E_i^* E_{p1})}{I_{sat}} \left(\frac{f_c}{k_B T_c} + \frac{f_h}{k_B T_h} \right), \quad (1)$$

where E_s (E_i) is the signal (idler) electric-field strength and $T_{c,h}$ are the temperatures along $\hat{G}$ (which are approximately equal to $T'_{c,h}$ for $\theta/2 \ll 1$). In our experiment, we find that $T_c \cong 2.5 \mu\text{K}$ and $T_h \cong 25 \mu\text{K}$ and that both are largely insensitive to I_p and Δ [24].

Substituting the atomic polarization given above into Maxwell's equations, we find that the steady-state coupled wave equations for the signal and idler beams become

$$\frac{dE_s}{dz} = i\kappa E_s + i\beta E_i^*, \quad \frac{dE_i^*}{dz} = i\kappa^* E_i^* + i\beta E_s, \quad (2)$$

where we make the constant-pump-beam approximation and assume that the optical fields adiabatically follow the evolution of the atomic density grating. We define the nonlinear coupling coefficient

$$\beta = \beta_c + \beta_h = \frac{\alpha_{\Delta=0} \hbar\Gamma}{64(\Delta/\Gamma)^2} \frac{I_p}{I_{sat}} \left(\frac{f_c}{k_B T_c} + \frac{f_h}{k_B T_h} \right), \quad (3)$$

such that βL corresponds to the nonlinear phase shift imposed on the weak probe beams by the interaction and $\kappa = \kappa_R + i\kappa_I = \beta + \alpha_{\Delta=0}(-2\Delta/\Gamma + i)/8(\Delta/\Gamma)^2$.

From eq. (2), we find that [36]

$$T = \frac{|w|^2}{|w \cos(wL) + \kappa_I \sin(wL)|}, \quad (4)$$

$$R = \frac{|\beta \sin(wL)|^2}{|w \cos(wL) + \kappa_I \sin(wL)|^2}, \quad (5)$$

where $w = (|\beta|^2 - \kappa_I^2)^{1/2}$. The solid lines in fig. 2 demonstrate that the predictions of eq. (5) fit the experimental data well over all measured values of Δ and I_p , where the reduced chi-squared value is $\chi_{red}^2 = 1.6$. To obtain the theoretical curves, we input the measured values of $\alpha_{\Delta=0}L$, Δ , $T_{c,h}$, and I_c into our model. This effectively extends our model beyond the simplified level structure considered theoretically and provides accurate quantitative predictions for our experimental configuration with no free parameters.

In order to quantify the nonlinear response, we use eq. (5) to determine β directly (see fig. 4(a)). For small pump intensities (*i.e.*, $I_p \ll I_c$), a Taylor series expansion of β about $I_p = 0$ yields

$$\beta \cong \frac{\alpha_{\Delta=0} \hbar\Gamma}{64(\Delta/\Gamma)^2} \frac{I_p}{I_{sat}} \left(\frac{I_p/I_c}{k_B T_c} + \frac{1 - I_p/I_c}{k_B T_h} \right). \quad (6)$$

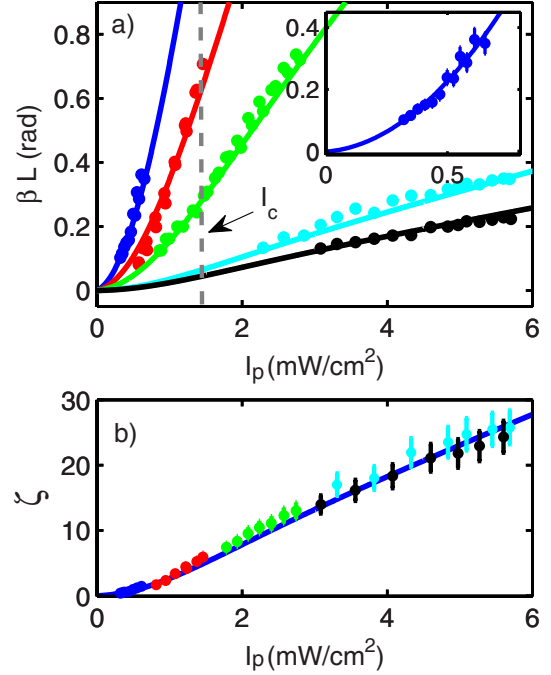


Fig. 4: (Color online) (a) Nonlinear phase shift corresponding to the data in fig. 2(a) obtained via eq. (4). The vertical dashed line indicates the value of the characteristic intensity I_c . The inset shows that, for $I_p \ll I_c$, β is well fit ($r^2 = 0.994$) by a quadratic function (where we show results for $\Delta/\Gamma = -3$). (b) Cross-phase modulation figure of merit as a function of I_p for the same detunings as in (a). The fact that all points follow the same curve indicates that both β and α_{Δ} scale as $(\Delta/\Gamma)^{-2}$. For both panels, $\alpha_{\Delta=0}L = 13.4(\pm 0.5)$, we use the $\mathbb{N}\perp$ configuration, the points correspond to the experimental data, and the solid curves correspond to the theory via eqs. (3) and (4).

The quadratic (linear) components of β correspond to a $\chi^{(5)}$ ($\chi^{(3)}$) response according to $\beta = |k_p|/(4\epsilon_0 c)[\chi_{xxyy}^{(3)}I_p + (2\epsilon_0 c)^{-1}\chi_{xxxxxy}^{(5)}I_p^2]$ for the polarizations shown in fig. 1. Using this relationship, we find $\chi_{xxxxxy}^{(5)} = 1.9(\pm 0.3) \times 10^{-12} (\text{m/V})^4$ for $\Delta = -3\Gamma$ (see inset in fig. 4(a)). This $\chi^{(5)}$ response is due almost entirely to the cold component (since $T_c/T_h = 0.1$) and gives rise to six-wave mixing (SWM), where the pump beams simultaneously undergo Bragg scattering and transfer atoms from the hot to the cold component (*i.e.*, cool the atoms). The value of $\chi^{(5)}$ is the largest ever reported, exceeding that obtained for EIT-based SWM by 10^7 [10,16] and in three-photon absorption in a zinc blend semiconductor by 10^{24} [37].

For the same detuning ($\Delta = -3\Gamma$), we find $\chi_{xxyy}^{(3)} = 2.0(\pm 2.3) \times 10^{-10} (\text{m/V})^2$, where the large experimental uncertainty arises from our inability to accurately measure β at the smallest I_p . This $\chi^{(3)}$ response is ~ 100 times smaller than that observed in EIT-based systems [6], and leads to four-wave mixing (FWM) in which the pump beams scatter off the weak atomic density grating composed solely of atoms in the hot component [38].

Beyond the region in which the Taylor series expansion given in eq. (6) is valid (*i.e.*, $I_c < I_p \ll I_{sat}^\Delta$), $f_c \cong 1$ and the $\chi^{(5)}$ response saturates. Nevertheless, β continues to increase linearly with I_p , in contrast to $\chi^{(3)}$ processes that increase only sub-linearly beyond saturation.

We compare our observed NLO response to previously reported nonlinearities based on $\chi^{(3)}$ processes by considering the achievable NLO phase shift for a fixed I_p . For example, Lo *et al.* [6] observe a 0.25 rad phase shift for an intensity of $230 \mu\text{W}/\text{cm}^2$. We find that $\beta L = 0.25$ rad for $I_p = 560 \mu\text{W}/\text{cm}^2$ and $\Delta/\Gamma = -3$, although our system does not require any auxiliary, strong coupling beams (as in the EIT-based setups). Also, β continues increasing quadratically with I_p for our $\chi^{(5)}$ process (rather than linearly as in $\chi^{(3)}$ processes), which further aides us in achieving large phase shifts at low light levels. Our system provides the additional advantage that we can work far from resonance and thereby combine large nonlinearities with high transparency (similar to recent predictions for Rydberg-enhanced EIT [5]). We quantify the tradeoff between absorptive loss and NLO phase shift using the cross-phase modulation figure of merit $\zeta \equiv \beta/\alpha_\Delta$ (*i.e.*, the ratio of the NLO phase shift to power loss, see fig. 4(b)). While Lo *et al.* [6] observe a maximum value of $\zeta = 0.35$ for an intensity of $230 \mu\text{W}/\text{cm}^2$, we exceed this value for $I_p > 300 \mu\text{W}/\text{cm}^2$ and obtain a maximum value of $\zeta = 26$ for $I_p = 5.6 \text{ mW}/\text{cm}^2$. Our system's ability to realize large ζ arises directly from the independence of I_c on detuning.

For sufficiently large values of βL , the back-action of the amplified probe fields influences the coupled light-matter dynamics, resulting in a strongly coupled system with long-range atom-atom interactions. In this regime, we observe additional atomic cooling in the (x, z) -plane and small group velocities of light on the order of $v_g \cong c/10^5$ [24]. The coherence time of the system also increases to $\sim 100 \mu\text{s}$, which is over 50 times larger than that measured for $\beta L \ll 1$, where thermal atomic motion causes grating washout. For $\beta L > 1$, we observe a collective instability in which the probe fields and density grating are self-generated via the NLO interaction [24,39].

While we focused on the $\aleph_\perp$ configuration in the preceding analysis, we observe qualitatively similar results for other polarization configurations. In particular, we find that the configurations in which the pump lattice supports atomic cooling and bunching (*i.e.*, $\aleph$, Θ , and Σ) all display a NLO response that increases quadratically (linearly) with I_p below (above) I_c . This scaling reinforces our claim that the formation of density gratings are integral to the nonlinearity, and indicates that the details of the level scheme are not critical to the physics involved.

In addition to studying how the nonlinearity scales with I_p , we also consider the dependence of the magnitude of the NLO response on the particular polarization configuration. Figure 5 shows that the nonlinear phase shift for a fixed pump intensity varies greatly with the choice of beam polarizations. We compare the configurations quantitatively by determining the slope of the nonlinear phase

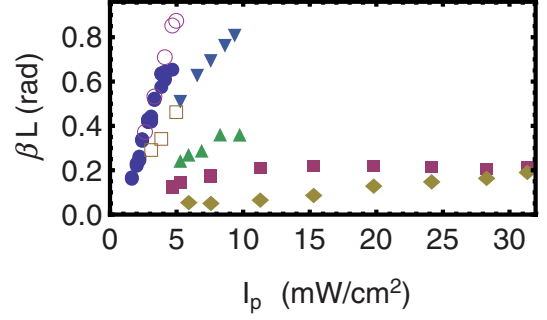


Fig. 5: (Color online) (a) Dependence of the nonlinear phase shift on I_p for the $\Theta|$ (solid circle), $\aleph_\perp$ (open circle), $\aleph_\parallel$ (open square), $\Sigma_\parallel$ (up-pointing triangle), $\Sigma_\perp$ (down-pointing triangle), $\Xi_\perp$ (solid square), $\Xi_\parallel$ (diamond). For all data, $\Delta/\Gamma = -7$ and $\alpha_{\Delta=0}L = 13.4(\pm 0.5)$.

Table 2: Polarization dependence of the slope of the nonlinear phase shift in units of (mrad/(mW/cm²)) for the data shown in fig. 5. Typical uncertainties are between 5% and 10% of the values given.

	$\aleph_\parallel$	$\aleph_\perp$	$\Sigma_\parallel$	$\Sigma_\perp$	$\Xi_\parallel$	$\Xi_\perp$	$\Theta_\parallel$
m	94	173	42	91	10	6	146

shift $m = \beta L/I_p$ in the region where $I_p > I_c$ (see table 2). The largest values of m correspond to the configurations discussed above, in which atomic localization in the pump lattice assists the formation of the pump-probe grating. In contrast, we observe significantly smaller values of m for the Ξ configurations, which cool but do not bunch the atoms. This indicates the importance of atomic bunching for realizing large nonlinearities in our system.

We also investigate the nonlinear coupling strength for blue detunings (*i.e.*, $\Delta > 0$), where we find that βL is ~ 100 times smaller than for red lattices with all other parameters equal. This is because red lattices lead to a net cooling and atomic localization at the intensity antinodes, whereas blue lattices result in atomic heating and localization in the intensity nodes. Thus, the atoms in the red-lattice experience a larger average electric-field strength, which more efficiently drives the nonlinearity for $I_p \ll I_{sat}$ [40]. In addition, the lower atomic temperatures in the red lattice lead to an enhanced density grating contrast and scattering efficiency. Thus, the NLO process occurs for any setup that permits both atomic cooling and bunching, which is consistent with our physical interpretation of the nonlinearity.

In conclusion, we observe experimentally a NLO mechanism based on Bragg scattering via an atomic density grating that is enhanced by dissipation in the gas' external degrees of freedom. At low light levels, this leads to the largest measured $\chi^{(5)}$ response and highly efficient SWM for a range of parameters over which the gas is highly transparent. We build on our previous work by providing

additional results that further back up our interpretation of the nonlinearity. For different polarization configurations, the light-matter coupling becomes strong enough that the generated optical fields act back on the gas and lead to interesting collective effects. Thus, this system represents a particularly interesting platform for studying collective, many-body physics [22], provides a novel *in situ* diagnostic tool for studying optical lattices [24], and may be useful in enhancing applications based on phase-sensitive amplification, nonclassical light generation, and interferometry. In addition, while the presented theoretical model describes well the observed steady-state experimental results, a complete description of the light-matter dynamics in the collective regime (*i.e.*, where the back-action of the probe fields becomes important) requires a nonperturbative approach. Such a model would necessarily combine previous work on atomic motion in dissipative, multidimensional optical lattices [28] with models of collective scattering via atomic motion in conservative lattices [25].

We gratefully acknowledge the financial support of the US NSF through Grant No. PHY-0855399.

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